

# HAROLD

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## EDUCATION

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### Holy Angel University Angeles City, Pampanga, Philippines

*Bachelor of Science in Computer Science, School of Computing* Expected 2025

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Mobile Development, Software Engineering
- Capstone Research: HandsFree — Offline Voice-Controlled Android App (Tagalog/English/Taglish ASR)

## RESEARCH & CAPSTONE

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### HandsFree: Offline Hands-Free Android Application Holy Angel University

*Lead Developer & Researcher* 2024 – Present

- Designed an offline overlay Android app capturing voice commands in Filipino, English, and Taglish with zero network dependency and zero data transmission.
- Architected a translation normalization layer, transformer-based ASR fine-tuning pipeline (OpenAI Whisper), and intent classifier dispatching simulated gestures via Android Accessibility Service API.
- Applied transfer learning via fine-tuning on Tagalog-English-Taglish corpus (Mozilla Common Voice, Google FLEURS, LibriSpeech) using cross-entropy loss and gradient-based optimization.
- Implemented noise suppression preprocessing (WebRTC/RNNoise) and confidence thresholding for out-of-vocabulary rejection.
- Delivered five core app functions: App Launching, Smart Search, Messaging, Accepting Calls, and Scheduling.

## PROJECTS

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### Luna — AI VTuber (Personal Project) In Development

*Solo Developer* 2024 – Present

- Building a real-time AI virtual streamer powered by an LLM/TTS/RAG stack with Twitch and TikTok Live integration.
- Architecting retrieval-augmented generation (RAG) memory layer and speech synthesis pipeline for interactive, personality-consistent live responses.

## ACTIVITIES & LEADERSHIP

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### Google Developer Group (GDG) HAU Holy Angel University Chapter

*Internal Officer* 2024 – Present

## TECHNICAL SKILLS

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**Languages:** Python, JavaScript/TypeScript, Java, Go, Kotlin

**AI / ML:** Transformer fine-tuning (Whisper/HuggingFace), transfer learning, RAG, TTS pipelines, scikit-learn

**Mobile:** Android (Accessibility Service API), Kotlin

**Tools & Platforms:** Git, GitHub, Google Colab, Node.js, LibreOffice, whisper.cpp, sherpa-onnx

**Concepts:** Data Structures & Algorithms (NeetCode 150), ML theory fundamentals, offline-first architecture

## **ADDITIONAL**

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**Languages:** Filipino (native), English (professional proficiency), Taglish

**Interests:** AI/ML Engineering, Speech Recognition, VTubing Technology, Competitive Programming